SOFTWARE REQUIREMENTS
SPECIFICATION

Sports Center Management System (SCMS)

Reverse-engineered from repository implementation
Version 1.1 | 11 November 2026
Local repository baseline

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Appendix A. Traceability and Evidence Boundaries

This revision aligns the use-case and data-model sections with the supplied Draw.io diagrams, while distinguishing diagram scope from repository implementation evidence.

I. Project Overview

1. Project Introduction

The Sports Center Management System (SCMS) is a web-oriented system for operating a sports center. Its backend coordinates accounts and permissions, sports and facility catalogs, Normal group-class enrollment, Advanced one-to-one registrations, schedules, invoices and payment verification, training records, attendance, notifications, support chat, and operational reporting.

2. Project Objectives

Maintain consistent records for members, staff, centers, sports, rooms, classes, sessions, entitlements, attendance and financial evidence.

Support Normal group classes and Advanced one-to-one instruction through distinct enrollment and entitlement workflows.

Protect financial and scheduling changes with authorization, ownership/center validation, transaction boundaries, conflict checks and idempotency where implemented.

Provide member self-service and staff operations through role-aware APIs; the current frontend deployment identity remains unverified.

Provide auditable reports, notifications and customer-support conversation handling.

3. Project Scope

Included in repository	Boundary / evidence limitation
Spring REST API, JPA, PostgreSQL schemas/migrations, authentication/RBAC, enrollment, booking, payments, training, attendance, reports, chat and notifications.	No live deployment, runtime environment, SLA, production data or current FE build was independently inspected.
A React/Vite UI exists in FE_Project-main.	AGENTS.md labels FE_Project-main an old reference; it is not asserted to be the currently integrated FE.
PostgreSQL/Supabase is the documented operational DB choice.	SQL Server scripts/driver exist; runtime acceptance is not established. External credentials and provider status are unknown.
D01-D19 record current business decisions.	A decided rule is not evidence that a cloud migration or live release passed.

4. Target Users

Roles are Admin, Center Manager, Receptionist, Coach and Member, plus unauthenticated visitors. Access combines role, dynamic authority, account state, center scope, ownership, assignment and business-state checks; a role name alone does not guarantee access. The supplied diagrams are design views: the use-case actor set and ERD actor taxonomy are not identical, so this SRS records their mismatch instead of treating the diagrams as a single normalized role list.

5. Main System Capabilities

Account registration, login/session restoration, Google sign-in integration, email OTP/password recovery, profile and avatar handling.

Catalog/resource administration; Normal class publishing, sessions, enrollment and waitlists.

Advanced registration, Coach acceptance, invoice, 12-credit entitlement and flexible private-session booking.

Cash/VNPay payment, invoice/receipt, QueryDR, reconciliation and payment resolutions.

Attendance, QR check-in, correction, training plans/results, evaluations and class feedback.

Holiday impact preview, center-incident recovery, notification delivery, support chat and reports.

II. Technical Overview

1. Technology Stack

Layer	Repository-verified components
Backend	Java 17; Spring Boot 3.3.5; Spring Web, Data JPA, Security, Validation, Mail, Actuator; springdoc OpenAPI 2.6.0; Lombok.
Frontend artifact	React 19, React Router 7, Vite 8, Tailwind CSS 4, Recharts 3, Motion and lucide-react in FE_Project-main; current deployment is not known.
Data	PostgreSQL and Microsoft SQL Server JDBC runtime drivers; H2 test dependency. PostgreSQL is current operational choice. JPA/Hibernate mappings.
Authentication	Spring Security session; custom current-account and first-login filters; role/authority method checks; BCrypt-compatible hashing and legacy verification.
Integrations	VNPay IPN/return/QueryDR; Google Identity; SMTP; avatar/media; Anthropic and DeepSeek chat-provider components. Live provider acceptance is not established.
Build	Maven wrapper for backend; npm/Vite scripts for frontend; deployment configuration exists but does not prove a live service.

2. Development Tools and Infrastructure

Maven and Vite are the project build systems. Repository guidance distinguishes a data-preserving root PostgreSQL schema from database/schema_postgres.sql, which rebuilds named SCMS tables and reference data. These scripts serve different purposes.

3. System Architecture Overview

Browser UI (React/Vite reference; current deployment unknown)

| HTTP + session cookie / JSON

v

Spring MVC Controllers -> Domain Services / transactions -> Spring Data JPA

| Spring Security |

+---v

PostgreSQL / Supabase (documented DB)

VNPay | Google Identity | SMTP | Media | AI chat APIs

The backend is a modular monolith organized under com.sportscenter. Controllers expose REST resources; services apply validation, ownership/center rules, persistence, transaction and audit behavior. Scheduled jobs process expiry, payment timeout, attendance lifecycle and notifications.

4. Database Technology

The PostgreSQL reconstruction schema describes 47 SCMS tables. Root schema is intended to upgrade while preserving IDs and business evidence. database/schema_postgres.sql rebuilds SCMS content when run. The live database was not queried; ERD descriptions refer to checked-in DDL and JPA.

5. External Services and Integrations

Integration	Observed capability	Boundary
VNPay	Create links, validate signed IPN/return, persist payment attempt provenance, query and reconcile.	Credentials/network and sandbox/live acceptance unknown.
Google Identity	Verify identity token and support account sign-in/linking.	Cloud migration and live login acceptance are separate.
SMTP	Email OTP, recovery and notification support.	Configuration does not prove delivery.
Media/avatar	Upload/avatar URL handling and optimization.	Provider/live status not identified.
AI chat	FAQ/provider dispatch, quota and handoff components.	Provider key and actual model availability unverified.

III. Software Requirement Specification

1. Product Overview

SCMS supports a center operating cycle from account creation and catalog discovery through class or Advanced enrollment, payment, scheduling, instruction, attendance and reporting. Normal enrollment is tied to a class and seats. Advanced registration is tied to a subject and Primary Coach and activates entitlement only after valid payment. Staff operate users, resources, schedules, payment evidence, training, exceptions and reporting within explicit authority and center scope.

1.1 System Boundaries

SCMS records invoice/payment state and can verify VNPay; approved refunds are performed outside SCMS and recorded with evidence.

The API is authoritative for identity, role, price snapshot, entitlement, expiry, credit and schedule/payment state.

Gateway, Google identity, SMTP, media and AI services are outside the database transaction boundary.

The reference UI does not establish current screen-to-API integration.

2. User Requirements

2.1 Actors

#	Actor	Description
1	Visitor / Guest	Unauthenticated person browsing public catalog and initiating account registration or login. The ERD labels this actor “Visitor/Guess”; this SRS uses Visitor / Guest.
2	Administrator (ADMIN)	System operator with explicit permissions; the ERD actor sketch names Administrator, but the supplied use-case diagram does not draw this actor.
3	Center Manager	Center-scoped operator managing staff, resources, classes, schedules, incidents, holidays, payment resolutions and reports when authorized.
4	Receptionist	Front-desk staff supporting member and invoice operations with required authority.
5	Coach	Instructional staff viewing assigned work, deciding eligible Advanced requests and recording instruction/attendance data.

#	Actor	Description
6	Member	Learner managing own profile, enrollment, invoices, entitlement, schedule, attendance and support conversation.
7	Authenticated User (abstract)	Generalized parent actor for the five authenticated roles: Admin, Center Manager, Receptionist, Coach and Member. Shared authenticated capabilities inherit from this actor.

2.2 Use Cases

2.2.1 Use Case Diagram(s)

The supplied UseCase.drawio.xml has no UC identifiers. The capability names, actors and use-case links in Section 2.2.2 are transcribed from that diagram; IDs UC-01–UC-30 are assigned in source-node order for SRS traceability only. The use-case view draws Center Manager, Coach, Member, Receptionist and external AI System, Notification Service and Payment Gateway. It omits Administrator, Visitor/Guest and abstract Authenticated User, which appear in the ERD actor page; this scope difference does not establish removal of those roles from implementation. The diagram has no explicit include/extend stereotypes. Relationship entries therefore record only unlabeled diagram connectors. The diagram names membership packages, while project decision D03 defines Normal/Advanced learning entitlements and retires the generic membership-package model; Section 4 preserves D03 and records this terminology as a design conflict.

2.2.2 Use Case Descriptions

ID	Use Case	Actors	Description	Relationship
UC-01	Manage Users	Center Manager	Center Manager manages users.	—
UC-02	Manage Classes & Schedule	Center Manager	Center Manager manages classes & schedule.	—
UC-03	Assign Coach to Class	Center Manager	The Center Manager assigns a Coach to a class.	—
UC-04	View Reports	Center Manager	Center Manager views reports.	—
UC-05	Manage Membership Packages	Center Manager	Center Manager manages membership packages.	—
UC-06	Manage System Permissions	Center Manager	Center Manager manages system permissions.	—
UC-07	View Teaching Schedule & Student List	Coach	Coach views teaching schedule & student list.	—
UC-08	Manage Training Plan	Coach	Coach manages training plan.	Unlabeled connector to UC-09 (Send Notifications/ Homework); Unlabeled connector to UC-12 (Use AI to suggest exercises)
UC-09	Send Notifications/ Homework	Coach; Notification Service (external)	The Coach sends notifications or homework.	Unlabeled connector to UC-08 (Manage Training Plan)
UC-10	Evaluate Student Progress	Coach	Coach evaluates student progress.	—
UC-11	Record Training Results & Attendance	Coach	Coach records training results & attendance.	—
UC-12	Use AI to suggest exercises	AI System (external)	The AI System supports exercise suggestions.	Unlabeled connector to UC-08 (Manage Training Plan)
UC-13	Manage Account	Member	Member manages account.	—
UC-14	Manage Class Registration	Member	Member manages class registration.	—
UC-15	View Personal Schedule & Coach Info	Member	Member views personal schedule & coach info.	—

ID	Use Case	Actors	Description	Relationship
UC-16	Consult AI Assistant	Member; AI System (external)	The Member consults the AI assistant.	Unlabeled connector to UC-29 (View Training Plan & Feedback)
UC-17	View Notifications	Member	Member views notifications.	—
UC-18	Register/Renew Membership Package for Member	Receptionist; Member	The Member and Receptionist use the system to register or renew a membership package.	Unlabeled connector to UC-19 (Process Payment & Invoice)
UC-19	Process Payment & Invoice	Receptionist; Payment Gateway (external)	The Member and Receptionist use the system for payment and invoice processing.	Unlabeled connector to UC-18 (Register /Renew Membership Package for Member)
UC-20	Manage Class Registration for Member	Receptionist	Receptionist manages class registration for member.	Unlabeled connector to UC-23 (Receive Support Requests)
UC-21	Manage Member Information	Receptionist	Receptionist manages member information.	—
UC-22	Take Member Attendance	Receptionist	Receptionist takes member attendance.	—
UC-23	Receive Support Requests	Receptionist	The Receptionist receives support requests.	Unlabeled connector to UC-20 (Manage Class Registration for Member)
UC-24	View System Logs	Center Manager	Center Manager views system logs.	—
UC-25	Register Account	Receptionist	Receptionist registers account.	—
UC-26	View Class List & Schedule	Member	Member views class list & schedule.	—
UC-27	View Student Profile & Goals	Coach	Coach views student profile & goals.	—
UC-28	View Training History & Results	Member	Member views training history & results.	—
UC-29	View Training Plan & Feedback	Member	The Member views a training plan and feedback.	Unlabeled connector to UC-16 (Consult AI Assistant)
UC-30	Check Member Package Status	Receptionist	The Receptionist checks a Member package status.	—

3. Functional Requirements

3.1 System Functional Overview

3.1.1 Screens Flow

Application / role	Route flow in FE_Project-main (reference only)
Visitor	Home/course/Coach views -> login/register -> role destination.
Admin	Login -> dashboard -> users/roles/audit/profile.
Manager	Dashboard -> staff/members/packages/classes/resources/payments/reports/audit.
Coach	Dashboard/schedule -> assigned members -> attendance -> progress/training.
Receptionist	Dashboard -> member search -> class booking -> payments.
Member	Dashboard -> membership/catalog -> attendance/progress -> profile/support.

FE_Project-main is marked an old reference by repository guidance. Its routes do not prove current integration or deployment.

3.1.2 Screen Descriptions

#	Feature	Screen	Description
1	Authentication	Login/Register	Credentials, registration, Google, recovery and session restoration.
2	Public catalog	Home / sport detail	Public landing and reduced subject discovery.
3	Admin	Dashboard	System summary/navigation in reference UI.
4	Admin	Users / roles / audit	Account, permission and audit functions behind authorities.
5	Manager	Dashboard	Center operation summary.
6	Manager	Staff / members	Search and manage center staff and members.
7	Manager	Classes / resources	Manage classes, sessions, rooms and feedback.
8	Manager	Payments / reports	Read payment and report data according to API scope.
9	Coach	Schedule / dashboard	Assigned teaching and class context.
10	Coach	Attendance	Roster, attendance and correction context.
11	Coach	Training / progress	Plans, results and learner evaluation.
12	Receptionist	Member search	Find or register member at front desk.
13	Receptionist	Class booking	Assist eligible Normal enrollment.
14	Receptionist	Payments	Cash/invoice operations with MANAGE_INVOICES.
15	Member	Dashboard / membership	Own enrollment/Advanced entitlement.
16	Member	Class catalog	Discover and enroll in Normal classes.
17	Member	Attendance / progress	Read own learning records.
18	Member	Profile / support	Own profile/avatar and support conversation.

3.1.3 Screen Authorization

Function	Visitor	Admin	Manager	Receptionist	Coach	Member
Public catalog	Y	Y	Y	Y	Y	Y
Own profile	—	Own	Own	Own	Own	Own
User/staff admin	—	Authority + target	Center authority	Limited authority	—	—
Class/resource admin	—	Authority	Authority + center	If granted	—	—
Normal enrollment	—	—	If granted	If granted	—	Own
Advanced decision/order	—	—	Support authority	—	Primary Coach	Own order
Private schedule	—	—	Support authority	—	Primary Coach	Own read
Invoice/payment	—	Scope authority	Scope authority	Invoice authority	—	Own invoice
Attendance/training	—	Admin authority	Manager authority	—	Assignment	Own read
Reports	—	VIEW_REPORTS	VIEW_REPORTS	—	—	—
Chat	—	HANDLE_CHAT	HANDLE_CHAT	If granted	If granted	CHAT_AS_MEMBER
Audit	—	VIEW_AUDIT_LOG	If granted	—	—	—

Capability summary only. Actual checks combine role, authority, active state, ownership, center, assignment and service business rules. See SecurityConfig, controller @PreAuthorize and domain services.

3.1.4 Non-Screen Functions

#	Feature	Function	Description
1	Invoice expiry	Scheduled job	Timeouts unpaid holds at configured time.
2	Entitlement expiry	Scheduled job	Expires Advanced entitlement and stores actual recovery facts.
3	Attendance lifecycle	Processor	Finalizes ended sessions and queues post-cutoff review idempotently.
4	Notification schedule	Job/events	Creates and sends due notifications after committed events.
5	Payment verification	Gateway/API	Checks signature/reference/amount/idempotency; Return does not set Paid.
6	Email	Async service	Sends OTP/recovery/business email via configured SMTP.
7	Chat responder	Provider dispatch	FAQ/provider answer, quota and staff handoff.
8	Migration audit	Database migration	Guarded migration markers support replay and audit.

3.1.5 Entity Relationship Diagram

The supplied erd.drawio (1).xml contains four pages: an actor generalization sketch, two data-model pages (the second is titled as a copy), and a context-diagram title page. The actor page shows Authenticated User (abstract) generalized above Member, Visitor/Guest, Receptionist, Coach, Administrator and Center Manager. The use-case diagram differs: it has no Administrator or Visitor/Guest actor and has external-service actors. Keep these distinctions visible when interpreting access requirements.

3.1.5.1 Conceptual ERD

Identity subtype structure: users 1:1 with Member, Coach, Receptionist or Center Manager profiles; role/permission assignment is modeled separately.

Center is the tenancy and operational boundary for users, rooms, classes and sessions.

Subject and Coach have a many-to-many qualification association through coach_specializations.

Classes contain sessions and class enrollments; Member enrollments refer to the class and Member.

Advanced registration, invoice/payment and Member entitlement form related but distinct business records.

An Advanced entitlement can authorize sessions; sessions relate to attendance and training results.

Sessions retain change, absence-notice and incident impact records.

Notifications and per-user read state are separated; support conversations contain messages and read-state records.

Roles and permissions use a bridge; invoices link payment attempts and resolution evidence.

3.1.5.2 Logical ERD

users(user_id PK) -> subtype tables (user_id PK/FK)

centers(center_id PK) -> users/rooms/classes/sessions.center_id

roles(role_id PK) -> users; role_permissions(role_id, permission_id composite PK)

classes(class_id PK) -> sessions/class_enrollments; subjects(subject_id PK) -> classes/member_packages

invoices(invoice_id PK) -> payment_resolutions/vnpay_payment_attempts.invoice_id

member_packages(subscription_id PK) -> sessions/subscription_pauses

sessions(session_id PK) -> attendances/training_results/absence notices/session changes

ERD is a core relationship view, not exhaustive column documentation. See checked-in PostgreSQL schemas for complete keys and constraints. The Draw.io data pages are the design reference; the adjacent entity inventory and key descriptions remain repository/schema observations and may expose implementation differences.

#	Entity	Purpose, primary key and important foreign keys
1	users	Identity; PK user_id; FK role_id, center_id.
2	members	Member subtype/profile; PK/FK user_id.
3	coaches	Coach subtype; PK/FK user_id.
4	receptionists	Receptionist subtype; PK/FK user_id.
5	center_managers	Manager subtype; PK/FK user_id.

#	Entity	Purpose, primary key and important foreign keys
6	centers	Tenant and scheduling mutex; PK center_id.
7	roles	Role definitions; PK role_id.
8	permissions	Authority catalog; PK permission_id.
9	role_permissions	Role-authority bridge; composite PK/FKs.
10	email_otps	Email OTP state associated with identity.
11	subjects	Sport metadata and prices; PK subject_id.
12	specializations	Coach qualification catalog; PK specialization_id.
13	coach_specializations	Coach/qualification bridge; FKs both sides.
14	rooms	Center resources; PK room_id; FK center_id.
15	classes	Normal class or Advanced template; PK class_id; subject/center links.
16	sessions	Concrete lesson; PK session_id; class/center/room/coach/subscription links.
17	class_enrollments	Normal enrollment/waitlist; class and member FKs.
18	normal_class_requests	Class creation request and requester context.
19	member_packages	Advanced entitlement; member, subject, Primary Coach and invoice references.
20	membership_packages	Retired legacy catalog kept structurally; migration16 empties and blocks writes.
21	advanced_registrations	Advanced request/decision/order; member, subject, coach; unique member/requestKey.
22	coach_work_schedules	Coach schedule/acceptance policy; coach relation.
23	coach_work_slots	Weekly slot children of Coach schedule.
24	coach_days_off	Coach unavailable dates.
25	subscription_pauses	Entitlement pause/recovery intervals; subscription FK/source key.
26	invoices	Financial document/frozen amount; member and order links.
27	vnpay_payment_attempts	Gateway attempt provenance; invoice FK RESTRICT.
28	payment_resolutions	Refund/cash-correction workflow; invoice FK and unique invoice/type.
29	attendances	Current/history attendance; session/member and optional notice links.
30	session_absence_notices	Member notice; session/member links and supersession history.
31	training_plans	Instruction plans authored for class/session context.
32	training_results	Per-session Member result and Coach provenance.
33	evaluations	Learner evaluation with author/member relation.
34	class_feedbacks	Member class feedback; class/member relation and duplicate guard.
35	holidays	Closure calendar and operational impacts.
36	center_incidents	Center incident/idempotency record; session relation.
37	incident_affected_members	Incident-to-member/session bridge.
38	session_changes	Schedule change/audit/idempotency record; session FK.
39	notifications	Event/message content.
40	notification_recipients	Notification/user read-state bridge.
41	chat_conversations	Support conversation and assignment.
42	chat_messages	Conversation message and sender.
43	chat_read_state	Per-user conversation read position.
44	ai_consultations	Future-facing structure; not proof of a separately shipped workout recommender.
45	support_requests	Future-facing support request structure.
46	system_logs	Audit events and actor/target provenance.

#	Entity	Purpose, primary key and important foreign keys
47	scms_data_migration_audit	Migration marker and guarded replay evidence.

4. Detailed Functional Specifications

Specifications use the reference report's Trigger, Actors, Data Processing, Screen Layout and Function Details pattern. A current FE screen is not claimed unless verified; API/resource is the interface where that is the available evidence.

4.1 Authentication and account security

Related UC: UC-25 (Register Account), UC-13 (Manage Account).

Function Trigger: User submits credentials, OTP, Google token or recovery request.

Function Description: Actors: Visitor; Authenticated User. Purpose: support the named business workflow.

Data Processing: Validate identity, account and first-login state; establish/invalidate session or return generic recovery response; persist security state.

Screen Layout: Login/register/recovery forms; current UI unknown.

Function Details / Traceability: AuthController, RegistrationService, GoogleAuthService, PasswordResetService.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. Session identity is server-side. Client role is not trusted. Inactive/blocked and first-login gates apply.

4.2 Profiles and account administration

Related UC: UC-13 (Manage Account), UC-01 (Manage Users), UC-21 (Manage Member Information), UC-06 (Manage System Permissions), UC-24 (View System Logs).

Function Trigger: User opens profile or authorized staff submits account mutation.

Function Description: Actors: Authenticated User for self; Admin/Manager/Receptionist per authority. Purpose: support the named business workflow.

Data Processing: Resolve actor from session; validate allowed fields; enforce role-target and center limits; persist profile/account state and audit.

Screen Layout: Profile/avatar, staff/account management forms, role-permission matrix, and audit log view in the repository reference UI; current screen/API integration is not verified.

Function Details / Traceability: MemberSelfProfileService, UserController, MemberController, StaffController, AvatarService, RoleController/RoleService and AuditController/AuditService. For UC-04, an authorized Admin (or Manager when granted) reads role/permission data, updates role-permission assignments through the RBAC interface, and reviews audit records under VIEW_AUDIT_LOG. The backend validates the target role and authority, persists permission changes, and records the acting principal. Member self-profile exposes seven allowed fields; role/status/center/package/credit/amount are excluded.

4.3 Catalog, Coach directory and rooms

Related UC: UC-26 (View Class List & Schedule), UC-02 (Manage Classes & Schedule), UC-03 (Assign Coach to Class), UC-15 (View Personal Schedule & Coach Info).

Function Trigger: Browse or authorized operator maintains sport, specialization or resource.

Function Description: Actors: Guest/authenticated reader; Manager/Admin with MANAGE_CLASSES. Purpose: support the named business workflow.

Data Processing: Project permitted read model; apply center/filter/contact policy; validate code/capacity and qualification; save resource changes.

Screen Layout: Public course and manager resource screens (reference).

Function Details / Traceability: PublicCatalogController, SubjectController, CoachDirectoryService, RoomController.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. Public catalog is reduced. Coach qualification is explicit. Directory uses transactional DTO/batch access to prevent lazy-load/N+1 issues.

4.4 Normal classes and enrollment

Related UC: UC-02 (Manage Classes & Schedule), UC-14 (Manage Class Registration), UC-20 (Manage Class Registration for Member), UC-26 (View Class List & Schedule).

Function Trigger: Staff publishes class/session or Member enrolls.

Function Description: Actors: Authorized staff; Member; supported receptionist. Purpose: support the named business workflow.

Data Processing: Check class state, center, active/block, duplicate, payment/entitlement, timetable and capacity; update enrollment/waitlist atomically.

Screen Layout: Class management, catalog and enrollment forms.

Function Details / Traceability: SportsClassController, EnrollmentController/Service, NormalClassCreationService.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. Advanced class enrollment is rejected and directed to Advanced registration workflow.

4.5 Advanced registration/order

Related UC: UC-18 (Register/Renew Membership Package for Member), UC-19 (Process Payment & Invoice), UC-30 (Check Member Package Status). Request/decision steps are not separate use cases in the diagram.

Function Trigger: Member requests sport/Coach; Coach decides; Member orders accepted request.

Function Description: Actors: Member; Primary Coach; Manager support. Purpose: support the named business workflow.

Data Processing: Validate qualification, schedule/capacity and center; record decision; snapshot Subject price; create invoice; activate entitlement only on valid Paid.

Screen Layout: Coach selection/request and invoice screens (reference).

Function Details / Traceability: AdvancedRegistrationController/Service, CoachCapacityService, InvoiceService.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. 12 credits; 45 days from Paid with policy closure adjustments. Idempotency keys prevent duplicate order. General membership product is retired. Advanced order includes UC-16 for the shared invoice/payment workflow; the order snapshots price, while the shared payment flow alone confirms valid Paid state and activates entitlement.

4.6 Private scheduling and incident recovery

Related UC: UC-02 (Manage Classes & Schedule), UC-03 (Assign Coach to Class), UC-07 (View Teaching Schedule & Student List), UC-15 (View Personal Schedule & Coach Info). Incident recovery is not shown as a separate diagram use case.

Function Trigger: Coach books, or Manager reschedules/handles incident.

Function Description: Actors: Primary Coach; authorized Manager. Purpose: support the named business workflow.

Data Processing: Under transaction/locks check entitlement, credit, notice, coach, room, holiday, deadline and conflicts; persist schedule, consume credit and audit; notify after commit.

Screen Layout: Schedule and incident screens (reference).

Function Details / Traceability: SessionController/Service, CenterIncidentService, NormalMakeupService.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. Retry does not consume again. Substitution does not change Primary Coach. Recovery restores actual recorded credits, not automatic 12.

4.7 Cancellation and absence

Related UC: UC-14 (Manage Class Registration), UC-20 (Manage Class Registration for Member), UC-22 (Take Member Attendance). Advanced cancellation and short-notice absence are not distinguished in the diagram.

Function Trigger: Member cancels or reports short-notice absence.

Function Description: Actors: Member; Coach observes actual attendance. Purpose: support the named business workflow.

Data Processing: At least 12h: release resource and return one original credit. Under 12h: commit notice and Absent attendance together; preserve superseded history after time change.

Screen Layout: Own schedule/cancel; Coach attendance (reference).

Function Details / Traceability: SessionAbsenceService, SessionService, AttendanceService.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. No retroactive notice or ordinary Normal cancellation. Coach actual result can supersede notice state.

4.8 Invoice, VNPay and payment resolution

Related UC: UC-18 (Register/Renew Membership Package for Member), UC-19 (Process Payment & Invoice).

Function Trigger: Member pays/verifies; gateway calls IPN; Receptionist confirms Cash; Manager records resolution.

Function Description: Actors: Member; Receptionist; Manager. Purpose: support the named business workflow.

Data Processing: Use frozen amount; validate method/deadline/signature/reference/amount/idempotency; QueryDR uses persisted attempt; shared Paid logic; resolution stores evidence.

Screen Layout: Member invoice/receipt and staff payment screens (reference).

Function Details / Traceability: InvoiceController/Service, VnPayGateway/VerificationService, PaymentResolutionController.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. Return never sets Paid. Late confirmation may require reconciliation and must not restore entitlement. Refund is outside SCMS.

4.9 Attendance, QR and review

Related UC: UC-11 (Record Training Results & Attendance), UC-22 (Take Member Attendance). QR check-in and correction are not shown as separate diagram use cases.

Function Trigger: Coach records during lesson; Member scans; authorized actor corrects; scheduler finalizes.

Function Description: Actors: Coach; Member; Manager correction. Purpose: support the named business workflow.

Data Processing: Check assignment/center/time window; validate QR signature/TTL/replay; persist attendance/result/audit; finalize and enqueue review once.

Screen Layout: Roster/attendance/correction screens (reference; current mapping unknown).

Function Details / Traceability: AttendanceController/Service, AttendanceFinalizationJob.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. Retain history. Corrections need reason. Sanction toggle defaults off. Correction and attendance entry use distinct eligibility.

4.10 Training and feedback

Related UC: UC-07 (View Teaching Schedule & Student List), UC-08 (Manage Training Plan), UC-10 (Evaluate Student Progress), UC-11 (Record Training Results & Attendance), UC-27 (View Student Profile & Goals), UC-28 (View Training History & Results), UC-29 (View Training Plan & Feedback), UC-12 (Use AI to suggest exercises).

Function Trigger: Coach creates plan/result/evaluation; Member reads or submits feedback.

Function Description: Actors: Coach; Member. Purpose: support the named business workflow.

Data Processing: Validate assignment, center, registration and state; persist author and target data under applicable duplicate constraints.

Screen Layout: Plan, progress, evaluation and feedback pages (reference).

Function Details / Traceability: TrainingPlan/Result/Evaluation and ClassFeedback controllers/services.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. Member access is scoped to own eligibility and class material.

4.11 Holidays, notifications, chat and reports

Related UC: UC-04 (View Reports), UC-09 (Send Notifications/ Homework), UC-17 (View Notifications), UC-23 (Receive Support Requests). Holiday management and support chat are not fully represented in the diagram.

Function Trigger: Manager previews closure; event creates notification; user chats or requests report.

Function Description: Actors: Manager; recipients; Member; HANDLE_CHAT staff; report authorities. Purpose: support the named business workflow.

Data Processing: Preview/fresh-token confirmation; persist notification/read state; enforce chat ownership/handler; scope reports and separate people from registrations and gross/refund/net amounts.

Screen Layout: Holiday, notification, support and report screens (reference).

Function Details / Traceability: HolidayImpactService, NotificationService/Jobs, ChatService, ReportService.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. No silent bulk schedule change. Provider-backed AI depends on configuration. Reports require VIEW_REPORTS and center scope.

4.12 Scheduled lifecycle processing

Related UC: Not shown in the use-case diagram (system function).

Function Trigger: Configured scheduler reaches due time.

Function Description: Actors: System (not a use case actor). Purpose: support the named business workflow.

Data Processing: Use injected Clock and Asia/Ho_Chi_Minh; select due records; transact idempotent state change/evidence; retry safely as supported.

Screen Layout: No user screen; observe changed state/audit.

Function Details / Traceability: PaymentExpirationJob, PackageExpirationJob, AttendanceFinalizationJob, NotificationScheduleJob.. Related JPA entities/repositories, checked-in DDL, docs01 API contract and Dxx decision document the detailed state, errors and constraints. Job class is not proof of a production run; deployed schema/config must match code.

5. Non-Functional Requirements

ID / attribute	Verified requirement or limitation
NFR-01 Security	Session authentication, method-level role/authority checks, first-login gate, principal-derived actor and ownership/center validation exist. No penetration test performed.
NFR-02 Data integrity	Transactions, DB constraints, idempotency and locks protect selected high-impact operations; coverage varies by flow.
NFR-03 Time	Relevant policies/jobs use injected Clock and Asia/Ho_Chi_Minh. Deployed clock drift is unmeasured.
NFR-04 Availability	No uptime, failover, RTO/RPO or production monitoring target identified.
NFR-05 Performance	Coach directory addresses batch fetching/N+1. No general latency/load objective identified.
NFR-06 Maintainability	Domain packages, Maven tests, OpenAPI and explicit migrations support maintenance; no score claimed.
NFR-07 Compatibility	Java 17/Spring Boot 3.3.5 pinned. PostgreSQL is operational direction; SQL Server runtime remains unaccepted.
NFR-08 Privacy	Self-profile field allowlist and contact/record scope controls exist. Retention/legal privacy policy is incomplete.
NFR-09 Usability	Role screens exist in reference React UI. Accessibility, usability study, responsive acceptance and current live UI are unverified.
NFR-10 Integration resilience	Some retry/fallback behavior exists for VNPay, mail, Google, media and AI. External SLAs are unspecified.
NFR-11 Scalability	No workload, data-volume or horizontal scaling target identified.
NFR-12 Auditability	System logs, actor/reason/consent/payment provenance and history exist in selected flows; completeness varies.

Appendix A. Traceability and Evidence Boundaries

Area	Evidence	Interpretation
Business policy	docs/04-DEBUG-NOTE.md, D01-D19	Decision source, not evidence of implementation/release.
API contract	docs/01-API-CONTRACT-FE-BE.md and OpenAPI snapshot	Published shape; source-generated Swagger is not independent policy validation.
Implementation	src/main/java/com/sportscenter/**	Actual code; authorization can span controller and service.
Database	database/README.md, database/schema_postgres.sql, root schema and migrations	Checked-in DDL, not live DB inspection.
Frontend	FE_Project-main/src/**	Old reference per AGENTS.md; current active FE must be identified.
Operational state	docs/00-MASTER-CHECKLIST.md Active and scoped QA records	Historical environment/SHA results; no tests/cloud calls were run for this SRS.
Repository identity	Local supplied checkout	Pasted request had a placeholder GitHub URL; analysis used the accessible local repository.

The code, decision log, contract and database scripts can differ; this document labels policy, implementation and release evidence separately. The task created documentation only. No application tests, cloud mutations, external provider calls or deployment checks were performed.